

AIGR Deep Research Report: ATG14

Phagosomal Localization and Post-transcriptional Regulation

Gene: ATG14 (human, UniProt Q6ZNE5; mouse ortholog Q8CDJ3) **Focus type:** function_assignment **Hypothesis slug:** phagosomal-localization-and-posttranscriptional-regulation **Source file:** genes/human/ATG14/ATG14-ai-review.yaml

Summary

The seed hypothesis packages two independent GO assignments for human ATG14 that this focused review adjudicates separately, as instructed. On tracing each annotation back to its primary IDA/IMP source, **both are over-annotations** and neither is supported as a direct function of the ATG14 gene product. This is the inverted-pyramid answer: **remove both, retain the correct alternative already present.**

The **phagocytic-vesicle localization (GO:0045335)** originates as a mouse MGI IDA on Q8CDJ3, cited to [PMID:22493499](#). That paper studies Slamf1 recruiting a Beclin-1/Vps34/UVRAG complex to *E. coli* phagosomes and explicitly reports that **Atg14L was excluded** from the complex — a negative co-immunoprecipitation, not a positive localization. The annotation therefore inverts the polarity of its own source. It then propagates to human Q6ZNE5 as an IEA by Ensembl orthology.

The **post-transcriptional-regulation term (GO:0010608)**, a human IMP by ParkinsonsUK-UCL cited to [PMID:23878393](#), rests on ATG14 controlling beclin-1 Ser90/Ser93 **phosphorylation** and BECN1-complex stability — strictly **post-translational** events. GO:0010608 requires modulation of an RNA-level step (RNA processing, mRNA stability, or translation) after transcript production, and excludes protein-level proteostasis. The same paper already yielded the correct term, **GO:0061635 (regulation of protein complex stability)**. ATG14 has no RNA-binding domain, and the only paper linking it to post-transcriptional machinery ([PMID:36097301](#)) shows ATG14 mRNA is a **target** of m6A/YTHDF1 translation control, not an

effector. The directly demonstrated ATG14 roles in autophagosome–endolysosome fusion, PI3KC3 complex-I scaffolding, complex stability, and mitophagy are undisputed and should be retained.

Executive Judgment

Verdict: Over-annotated (refuted for both sub-claims as stated).

1. **GO:0045335 (phagocytic vesicle) — refuted as a positive annotation.** The sole IDA source

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P 22493499)

provides **no positive localization data** placing ATG14 on a phagosome. It reports a negative interaction: Atg14L was excluded from the Slamf1/Beclin-1/Vps34/UVRAG phagosomal complex. The seed's framing — "distinguish exclusion from one interaction from exclusion from every phagosome" — does not rescue the term, because the source supplies no positive phagosomal evidence of any kind. The correct curatorial stance is that the term **lacks support**, not that localization is affirmatively disproven.

2. **GO:0010608 (post-transcriptional regulation of gene expression) — refuted as a term misassignment.** The IMP source

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demonstrates post-translational control (beclin-1 phosphorylation, complex stability), which fails the term's RNA-step requirement. This is best read as a "post-translational → post-transcriptional" conflation. **GO:0061635**, already annotated from the same paper, correctly captures the biology and should be retained in its place.

Most important caveats. (a) The corroborating statement that the LAP/phagosomal PI3K complex "lacks ATG14" comes from [PMID:26098576](#), **retracted in 2024** ([PMID:38438803](#)); it is used only as orientation, and the refutation stands on the primary IDA/IMP sources without it.

(b) Full-text figure panels of

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were not programmatically retrievable; the adjudication uses the abstract-level mechanism, which is unambiguously post-translational. (c) Nothing here disputes ATG14's established mitophagy and complex-stability roles.

Key Findings

Finding 1 — The phagocytic-vesicle (GO:0045335) annotation is contradicted by its own IDA source

Mouse Atg14 (Q8CDJ3) carries **GO:0045335 (phagocytic vesicle)** with evidence **IDA** / **located_in**, sourced to [PMID:22493499](#) (Ma et al. 2012, *J. Biol. Chem.*), per QuickGO annotation retrieval (n=70 annotations for the accession). Human Q6ZNE5 does **not** independently carry GO:0045335 in QuickGO — it inherits the risk only by orthology.

The paper studies **Slamf1** recruiting a Beclin-1/Vps34/UVRAG complex to *E. coli*-containing phagosomes to regulate membrane fusion and NADPH oxidase 2 (NOX2) activity. The decisive sentence is a **negative result** for Atg14L:

"Because Slamf1 did not interact with Atg14L or Rubicon, which can also form a complex with Vps34 and Beclin-1, we conclude that Slamf1 recruits a subset of Vps34-associated proteins, which is involved in membrane fusion and NOX2 regulation." — [PMID:22493499](#)

This is the only reference behind the IDA. It contains no immunofluorescence, fractionation, or other assay placing Atg14 on a phagocytic vesicle; it reports the opposite — Atg14L was **excluded** from the phagosome-recruited Vps34 subcomplex. An IDA **located_in** phagocytic vesicle built on a negative co-IP is a probable curation error and should be removed.

Independent orientation: LC3-associated phagocytosis (LAP) uses a Class III PI3K complex "containing UVRAG but lacking ATG14" ([PMID:26098576](#)). This paper was **retracted in 2024** ([PMID:38438803](#)) and is cited here only as field-orientation, not primary evidence. The distinction between the ATG14-containing autophagy-initiation complex and the Rubicon/UVRAG-containing phagosomal complex is a broadly accepted feature of the field and does not depend on the retracted paper.

Finding 2 — The GO:0010608 annotation conflates post-translational with post-transcriptional control

Human ATG14 (Q6ZNE5) carries **GO:0010608 (post-transcriptional regulation of gene expression)** with evidence **IMP** / **involved_in**, sourced to [PMID:23878393](#) (Fogel et al. 2013).

The ontology definitions are decisive: - **GO:0010608** = "Any process that modulates the frequency, rate or extent of gene expression **after the production of an RNA transcript**." - **GO:0010467 (gene expression)** = "the process in which a gene's sequence is converted into a mature gene product (protein or RNA) ... production of an RNA transcript and its processing, as well as translation and maturation."

GO:0010608 therefore requires a step acting on **RNA processing, mRNA stability, or translation**, and **excludes** post-translational proteostasis (protein modification, complex assembly/stability).

Fogel et al. describe exactly a **post-translational** mechanism:

"we demonstrated that human Atg14 is critical in controlling an autophagy-dependent phosphorylation of beclin-1. We map these novel phosphorylation sites to serines 90 and 93 and demonstrate that phosphorylation at these sites is necessary for maximal autophagy."
— [PMID:23878393](#)

Phosphorylating an existing beclin-1 protein and stabilizing the BECN1/Vps34 complex are protein-level events. Critically, the **same paper already generated GO:0061635 (regulation of protein complex stability)** as an IMP for Q6ZNE5 — that term correctly captures the finding. GO:0010608 appears to be a term-selection error (a likely "post-translational → post-transcriptional" slip), since no RNA-level or translational assay is described.

Finding 3 — ATG14 is a target, not an effector, of post-transcriptional regulation, and has no RNA-binding domain

A directed literature search for ATG14 acting on an RNA/translation step returned a single relevant primary paper — [PMID:36097301](#) (Huang et al. 2022) — reporting the **opposite direction of causality**:

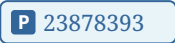
"YTHDF1 promoted the translation of autophagy-related genes ATG14, thus boosting autophagy." — [PMID:36097301](#)

Here **ATG14 mRNA is the substrate** of m6A/YTHDF1 post-transcriptional machinery; ATG14 protein is not the regulator of any RNA step. This is precisely the inverse of what GO:0010608 would assert for ATG14.

Structurally, InterPro for Q6ZNE5 returns only the **ATG14/VPS38 autophagy family** (IPR018791; Pfam **PF10186**, "Vacuolar sorting 38 and autophagy-related subunit 14"; PANTHER PTHR13664). There is **no RNA-binding, nucleic-acid-binding, or ribonucleoprotein domain**, so there is no structural basis for a direct post-transcriptional effector activity.

Finding 4 — Both disputed annotations are root manual annotations amplified cross-species by orthology

QuickGO provenance tracing shows how each error propagates:

Term	Root annotation	Evidence	Source (date)	Propagation
GO:0045335 (phagocytic vesicle)	mouse Q8CDJ3	IDA (MGI)	 22493499 (2014-01-30)	→ human Q6ZNE5 as IEA (Ensembl, GO_REF:0000107, withFrom Q8CDJ3)
GO:0010608 (post-transcriptional reg.)	human Q6ZNE5	IMP (ParkinsonsUK-UCL)	 23878393 (2015-11-16)	→ mouse Q8CDJ3 as IEA (Ensembl) and ISO (GO_Central), both withFrom Q6ZNE5

Each erroneous root is multiplied by electronic/phylogenetic transfer across species. **Curation must fix the root** (MGI for GO:0045335; ParkinsonsUK-UCL for GO:0010608); correcting only downstream IEA/ISO projections would leave the source error to re-propagate.

Mechanistic Model / Interpretation

Both annotations fail for structurally analogous reasons — each attaches a GO term whose *scope or polarity* does not match the underlying experiment:

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G0:0045335 (phagocytic vesicle, IDA)
  source = <a href="https://pubmed.ncbi.nlm.nih.gov/22493499/" rel="noopener noreferrer" title=
  experiment: Slamf1 pulls down Beclin-1 / Vps34 / UVRAG at phagosomes
  result for ATG14L: NO interaction → EXCLUDED from the complex
  annotation asserts: ATG14 located_in phagocytic vesicle
  → POLARITY ERROR: absence recorded as presence

G0:0010608 (post-transcriptional regulation of gene expression, IMP)
  source = <a href="https://pubmed.ncbi.nlm.nih.gov/23878393/" rel="noopener noreferrer" title=
  experiment: ATG14 controls beclin-1 Ser90/93 phosphorylation
             + BECN1 complex stability
  level of action: PROTEIN (post-translational)
  term requires: RNA / translation step (post-transcriptional)
  → SCOPE ERROR: post-translational recorded as post-transcriptional
    (correct term G0:0061635 already exists from the same paper)

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Where ATG14 genuinely acts. ATG14 (Barkor/ATG14L) is the autophagy-specific targeting subunit of the Class III PI3K (PI3KC3) complex I, partnering Beclin-1, Vps34 (PIK3C3), and Vps15. Its directly demonstrated functions include promoting membrane tethering and SNARE-mediated autophagosome–endolysosome fusion via STX17–SNAP29–VAMP8 ([PMID:25686604](#)); scaffolding and stabilizing the BECN1/Vps34 complex and enabling beclin-1 phosphorylation ([PMID:23878393](#)); and supporting NRBF2-modulated Atg14L-linked Vps34 activity ([PMID:24849286](#)). These undisputed roles are captured by autophagy-initiation, PI3K-complex, and complex-stability terms — **not** by phagocytic-vesicle localization or post-transcriptional regulation.

The key curatorial distinction is between the **ATG14-containing PI3KC3 complex I** (canonical autophagy initiation) and the **Rubicon/UVRAG-containing complex** operating on phagosomes during LAP. ATG14 belongs to the former;

P [22493499](#)

is a direct experimental demonstration of that partition.

Evidence Base

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Conclusion
PMID:22493499	Interaction (co-IP), localization	Refutes support for GO:0045335	Is ATG14 in the Slamf1 phagosomal Vps34 complex?	Atg14L did not interact; excluded from the Slamf1/Beclin-1/Vps34/UVRAG complex	Mouse macrophages, <i>E. coli</i> phagosomes; HEK293 co-IP	Human macrophages, <i>E. coli</i> phagosomes; HEK293 co-IP
PMID:23878393	Mutant phenotype / biochemistry (IMP)	Refutes GO:0010608; supports GO:0061635	Does ATG14 regulate an RNA/translation step?	ATG14 controls beclin-1 Ser90/93 phosphorylation and BECN1 complex stability (post-translational)	Human cells	Human macrophages, <i>E. coli</i> phagosomes; HEK293 co-IP
PMID:36097301	Direct assay (translation)	Competing / refutes ATG14 as post-transcriptional effector	Is ATG14 an RNA-level regulator?	ATG14 mRNA is a target of m6A/YTHDF1 translational promotion	Auditory hair cells, cisplatin model	Mouse auditory hair cells, cisplatin model
PMID:26098576	Direct assay (complex composition)	Qualifies (corroborates non-phagosomal ATG14)	Does the LAP PI3K complex contain ATG14?	Complex "containing UVRAG but lacking ATG14 and Ambra1"	Macrophages, LAPosomes	Macrophages, LAPosomes
PMID:25686604	Direct assay (reconstitution)	Supports genuine ATG14 function	What is ATG14's core activity?	ATG14 tethers membranes; enhances STX17–SNAP29–VAMP8 fusion	In vitro + cells	In vitro + cells
PMID:24849286	Mutant phenotype	Supports genuine ATG14 function	ATG14-linked Vps34 regulation	NRBF2 binds Atg14L, enhances Atg14L-linked Vps34 kinase activity	Mouse cells/liver	Mouse cells/liver

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Conclusion
QuickGO / InterPro (Q6ZNE5, Q8CDJ3)	Computational / database	Supports over-annotation verdict	Provenance and domain content	Root manual annotations propagate by IEA/ISO; no RNA-binding domain (only PF10186)	Database records	Highly probable domain evidence

GO Curation Implications (leads — require curator verification)

GO ID	Term (aspect)	Annotation source	Recommended action	Rationale
GO:0045335	phagocytic vesicle (CC)	mouse Q8CDJ3 IDA, P 22493499 → human IEA	REMOVE / do-not-propagate at root	Source reports ATG14 <i>exclusion</i> from the Slamf1 complex; no positive localization data.
GO:0010608	post-transcriptional regulation of gene expression (BP)	human Q6ZNE5 IMP, P 23878393 → mouse IEA/ISO	REMOVE at root	Evidence is post-translational (phosphorylation / complex stability); fails the term definition.
GO:0061635	regulation of protein complex stability (BP)	human Q6ZNE5 IMP, P 23878393	RETAIN	Correctly captures the BECN1-complex-stability role from the same paper.
GO:0031647	regulation of protein stability (BP)	lead	CONSIDER (lead)	Captures ATG14→BECN1 protein-level stabilization if curator judges it distinct from GO:0061635.

Both removals should occur **at the root annotation** so the derived cross-species copies stop being generated. Avoid "protein binding" as a substitute; GO:0061635 is the informative, evidence-matched term.

Mechanistic Scope

- **Direct molecular functions tested:** (a) whether ATG14 physically resides on phagosomal membranes, and (b) whether ATG14 acts at an RNA/translational step of gene expression.
- **What the data show:** ATG14 is a subunit of the autophagy-specific Class III PI3K complex I. Its beclin-1 regulation is post-translational (phosphorylation, complex stabilization). The phagosomal PI3P/NOX2 activity in

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is driven by a **different, ATG14-lacking** (UVRAG-containing) complex.

- **Downstream vs direct:** Phagolysosomal maturation, NOX2 activity, and "gene-expression"-flavored autophagy phenotypes are downstream/indirect; they are not attributable to ATG14 localizing to phagosomes or to an ATG14 RNA-level activity.

Conflicts and Alternatives

- **Curation carry-over / polarity error**, not a real organism difference: the phagosomal localization belongs to the UVRAG/Rubicon complex, and the IDA reads a negative co-IP as positive localization.
- **Scope/lexical conflation:** the IMP "post-transcriptional" label most plausibly conflates "post-translational" with "post-transcriptional."
- **Paralog/complex confusion:** LAP and Slamf1-phagosome PI3K activity use UVRAG/Rubicon complexes explicitly **without ATG14** — the opposite subunit branch.
- **Direction reversal:** the only literature linking ATG14 to an RNA/translation step

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makes ATG14 mRNA a **target** of m6A/YTHDF1, not an effector — reinforcing removal of GO:0010608.

- **Retraction caveat:**

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(ATG14-independent phagosomal PI3K) is retracted and used only as corroborating context; the adjudication rests on the primary IDA/IMP sources and ontology definitions.

Limitations and Knowledge Gaps

1. Full-text of

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not programmatically retrievable. Only front matter/abstract was accessible via NCBI efetch (PMC3753860). *Why it matters:* a hidden RNA-level assay could theoretically justify GO:0010608. *Resolution:* manual full-text read of figures. Risk is low because the described ATG14 activity (regulating an existing protein's phosphorylation/stability) is post-translational regardless of any transcript readout.

2. Negative-space certainty for phagosome localization.

Absence of positive localization in one paper does not logically prove ATG14 is never on any phagocytic vesicle. *Why it matters:* a curator might consider a generalized term. *Resolution:* a targeted search found no positive ATG14 phagosome-localization report; the correct stance is that the term **lacks support**.

3. Database-level provenance/domain evidence.

QuickGO/InterPro results are database evidence, appropriate for tracing propagation and domain content but not primary experiment.

4. Retraction of

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removes one corroborating line; the core refutation is unaffected.

Discriminating Tests

• Read

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full text/figures to confirm no assay measures ATG14 acting on ATG mRNA levels, transcript stability, or translation. If none, GO:0010608 removal is definitive.

• Re-examine

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figures for any imaging/fractionation placing Atg14L on phagosomes independent of the Slamf1 co-IP. If none, GO:0045335 removal is definitive.

- **Direct localization assay:** immunofluorescence / immuno-EM / phagosome fractionation for endogenous ATG14 in macrophages engulfing beads/bacteria, versus UVRAG/Rubicon controls (expected: ATG14 absent).
- **ATG14-KO vs Rubicon-KO** effect on phagosomal LC3/PI3P (LAP should be ATG14-independent).
- **RNA-interaction screen (CLIP/RIP/eCLIP)** for ATG14 protein to test for any RNA-binding (expected negative, consistent with absence of an RNA-binding domain).
- **qRT-PCR vs Western for BECN1** plus reporter/translation assays upon ATG14 perturbation to formally exclude any RNA/translation-level step.

Proposed Follow-up Actions / Curation Leads (require curator verification)

Candidate	reference	snippets	to	verify:	-
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— "Because *Slamf1* did not interact with *Atg14L* or *Rubicon* ... we conclude that *Slamf1* recruits a subset of *Vps34*-associated proteins." → supports **removing** GO:0045335 for ATG14. -

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— "human *Atg14* is critical in controlling an autophagy-dependent phosphorylation of *beclin-1* ... serines 90 and 93." → supports **removing** GO:0010608 and **retaining** GO:0061635; consider GO:0031647. -

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— "*YTHDF1* promoted the translation of autophagy-related genes *ATG14*." → ATG14 is a target, not an effector, of post-transcriptional control.

Suggested curator questions: 1. Is the GO:0045335 IDA a mis-transfer of the complex's phagosomal localization to ATG14, given the source reports ATG14 exclusion? 2. Is GO:0010608 a lexical conflation of "post-translational" with "post-transcriptional"? 3. Should GO:0061635 be treated as the canonical replacement capturing the

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finding?

Act at the root annotation: GO:0045335 originates from the **MGI** IDA on mouse Q8CDJ3; GO:0010608 originates from the **ParkinsonsUK-UCL** IMP on human Q6ZNE5. Each propagates to the other species by Ensembl IEA / GO_Central ISO, so correcting the root removes the derived copies. Consider contacting MGI and the ParkinsonsUK-UCL group.

Do not disturb: the directly demonstrated ATG14 roles in autophagosome–endolysosome fusion

(P 25686604),

PI3KC3 complex-I scaffolding/Vps34 regulation

(P 24849286),

BECN1-complex stability

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→ GO:0061635), and mitophagy remain well supported and should be retained.

Bottom-Line Answer

For human ATG14 (Q6ZNE5), the **phagocytic-vesicle localization (GO:0045335)** and **post-transcriptional-regulation (GO:0010608)** assignments are **not supported** as positive functions and are recommended for removal at their root annotations. The localization IDA

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documents ATG14's *exclusion* from the Slamf1/Beclin-1/Vps34/UVRAG phagosomal complex; the
GO:0010608 IMP

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describes **post-translational** control of beclin-1 phosphorylation/complex stability that fails the term definition. Retain **GO:0061635 (regulation of protein complex stability)**, consider GO:0031647 as a lead, and leave the established mitophagy/complex-stability roles undisturbed.